THE NOVA RESEARCH CURRICULUM

Mathematics

Grade 5 · Term 2

Student edition

Outcomes: MA-5.3 all four operations with fractions, MA-5.4 decimals, MA-5.5 percentages.

Lesson notes

MA-5.3 Fractions (Weeks 1–4)

Add and subtract unlike fractions and mixed numbers (convert to improper fractions, find the LCM of denominators). Multiply: numerator × numerator, denominator × denominator, then simplify. Divide: multiply by the reciprocal (3/4 ÷ 1/2 = 3/4 × 2/1). Use area models before rules.

MA-5.4 Decimals (Weeks 5–8)

Compare decimals by place value; convert fractions to decimals by division (3/8 = 0.375); multiply decimals (count decimal places); divide by a decimal by multiplying both numbers by 10 or 100.

MA-5.5 Percentages (Weeks 9–12)

Percent means "out of 100" (a 10 × 10 grid). Convert between percentages, fractions and decimals; find a percentage of a quantity; simple discount problems from shops in Pakistan.

▶ **Try it on NovaSchool** (open **Simulations** on your dashboard):

- **Percentages on a hundred square** — Percent means "out of 100". Colour squares and see the fraction, decimal and percentage.

Workbook pages

Page	Outcome	Activity
1–3	MA-5.3	Area models; mixed-number sums; multiplication and division.
4–5	MA-5.4	Compare and order; fraction-to-decimal; decimal multiplication and division.
6–7	MA-5.5	100-square shading; conversions table; sale-price problems.

Question bank

#	Outcome	Question	Your answer
1	MA-5.3	2/3 × 3/4 =	5/7 / 1/2 / 6/7 / 8/9
2	MA-5.3	5/6 − 1/4 =	4/2 / 7/12 / 1/2 / 4/10
3	MA-5.3	3/4 ÷ 1/2 =	3/8 / 1 1/2 / 2/3 / 3/2 of 1/2
4	MA-5.3	1 1/2 + 2 2/3 =	3 3/5 / 4 1/6 / 3 1/6 / 4
5	MA-5.3	2 1/4 × 2 =	4 1/4 / 4 1/2 / 4 1/8 / 2 1/2

#	Outcome	Question	Your answer
6	MA-5.4	3/8 as a decimal is …	0.38 / 0.375 / 3.8 / 0.83
7	MA-5.4	Which is greater: 4.56 or 4.6?	4.6 / 4.56 / equal
8	MA-5.4	2.5 × 1.2 =	2.7 / 3.0 / 30 / 0.30
9	MA-5.4	7.2 ÷ 0.9 =	0.8 / 8 / 80 / 6.3
10	MA-5.4	0.4 × 0.3 =	1.2 / 0.12 / 0.012 / 12
11	MA-5.5	35% as a fraction in lowest terms is …	35/10 / 7/20 / 3/5 / 1/35
12	MA-5.5	3/5 = ___%	35 / 60 / 30 / 53
13	MA-5.5	25% of 360 =	25 / 36 / 90 / 120
14	MA-5.5	15% of 200 =	15 / 30 / 45 / 185
15	MA-5.5	A Rs 1,500 shirt has 20% off. What is the sale price?	_____